

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

INPUT:

```
%{
#include <stdio.h>
#include <stdlib.h>
float sum = 0.0;
int count = 0;
int expected_count = 0;
%}
%option noyywrap
%%
[0-9]+(.[0-9]+)? {
if (count < expected_count) {
sum += atof(yytext);
count++;
if (count == expected_count) {
float average = sum / expected_count;
printf("\n--- Final Result ---\n");
printf("Expected Terms: %d\n", expected_count);
printf("Total Sum: %.2f\n", sum);
printf("Calculated Average: %.4f\n", average);
printf("-----\n");
exit(0);
}
}
}
.\n ;
%%

int main(void) {
```

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```
printf("Enter the number of terms (N) you wish to average: ");
if (scanf("%d", &expected_count) != 1 || expected_count <= 0) {
    fprintf(stderr, "Invalid number of terms entered. Exiting.\n");
    return 1;
}
while (getchar() != '\n');
printf("Enter the %d numbers separated by spaces or newlines: \n",
expected_count);
yylex();
if (count < expected_count) {
    printf("\nWarning: Only %d out of %d numbers were processed before input
ended.\n", count, expected_count);
}
return 0;
}
```

OUTPUT:

```
vboxuser@Ubuntu-05:~/Desktop/22co05 CD$ lex exp5.l
vboxuser@Ubuntu-05:~/Desktop/22co05 CD$ gcc lex.yy.c -o exp5 -lfl
vboxuser@Ubuntu-05:~/Desktop/22co05 CD$ ./exp5
--- LEX Fixed-Term Average Calculator ---
Enter the number of terms (N) you wish to average: 4
Enter the 4 numbers separated by spaces or newlines:
56
78
34
12

--- Final Result ---
Expected Terms: 4
Total Sum: 180.00
Calculated Average: 45.0000
```

CONCLUSION: LEX program to include line numbers in a given source file was implemented and executed successfully.